

Juniper Networks[®] vMX Virtual Router Release Notes Release Notes

Release 18.4R1
21 July 2020

These release notes accompany the Junos OS Release 18.4R1 of the Juniper Networks[®] virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

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Introduction

The vMX Virtual Router (vMX) is an MX Series router optimized to run on x86 servers. vMX is supported on Red Hat Enterprise Linux, CentOS, and Ubuntu.

NOTE: See “[System Requirements](#)” on page 5 for details on each supported environment.

vMX enables you to leverage Junos OS Release 18.4R1 to provide a quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations as the management and configuration are the same as for MX Series routers
- Introduces new services without requiring a reconfiguration of the current infrastructure

This release of vMX supports most of the features available on Juniper Networks MX Series routers.

Starting with Junos OS Release 18.4R1, vMX documentation has changed to a release-independent presentation model. This means that you will find only one version of vMX documentation in the TechLibrary pages on <https://www.juniper.net>. You can now view what has changed for individual features by release within the documentation pages. In addition, a summary of all changes made to the contents is provided in the Release History table at the end of the documentation page. To find details about the changes, use the links provided in the Release History table to navigate to the relevant location in the page.

New and Changed Features

This section describes the new features and enhancements in this release.

- Starting in Junos OS Release 18.4R1 or later, virtual forwarding plane (VFP) runs on Wind River 9.
- Starting in Junos OS Release 18.4, vMX adds support for multiqueue for the DPDK-based vrouter.

OpenStack Liberty supports the ability to create VMs with multiple queues on their virtio interfaces. Virtio is a Linux platform for I/O virtualization, providing a common set of I/O virtualization drivers. Multiqueue virtio is an approach that enables the processing of packet sending and receiving to be scaled to the number of available virtual CPUs (vCPUs) of a guest, through the use of multiple queues.

The number of queues is equal to the number of vCPUs defined for the instance. Two (2) queues per virtio NICs are supported.

Multi-queue virtio improves network performance by allowing parallel packet processing by creating multiple TX and RX queues.

- Starting in Junos OS Release 18.4R1 or later, virtual forwarding plane (VFP) runs on Wind River 9.
- **Unmodified IXGBE driver and i40e driver support**—Starting in Junos OS Release 18.4R1, vMX instances can be deployed with an unmodified IXGBE driver and i40e driver on Ubuntu version 16.04. Before installing the vMX instance, configure the unmodified IXGBE driver or the i40e driver. On a vMX instance, you are allowed to configure multiple virtual functions on the same physical port.
- **Weighted round-robin (WRR) of subscriber traffic on a port**—Starting in Junos OS Release 18.4R1, the vMX instances supports scheduling priority in QoS that can be set according to the weighted round-robin (WRR) algorithm. The scheduling priority determines the order in which an output interface transmits traffic from the queues to ensure that queues containing important traffic are provided better access to the outgoing interface.

Known Behavior

This section contains the known behaviors and limitations in this release.

Weighted Round-Robin of Subscriber Traffic on a Port Limitations

The following list describes the limitations for WRR:

- The delay-buffer rate must be configured for WRR to work correctly.
- A discrepancy in the delay-buffer rate values among the VLANs belonging to the same level 2 scheduler node can cause the WRR to work incorrectly.
- WRR does not work correctly if the ratio of the shaping rate is greater than 100 among all the subscribers.
- The number of level 2 scheduler nodes and the number of subscribers per level 2 scheduler node must be equal to 32,000 for it to work correctly.
- Any modification to the level 2 scheduler node configuration requires an FPC reset.

Known Issues

This section lists the known issues in this release.

- The input **int** field of the MPLS version 4 data records reports the SNMP index value of the LSI interface instead of the ingress physical interface. [PR1312047](#)
- The TX driver issue is detected on VF 0 and PF reset is issued. [PR1382787](#)
- The virtio throughput is the same for multi-queue and single-queue. [PR1389338](#)

Resolved Issues

This section lists the issues fixed in this release.

- While using the IXGBE driver with SRIOV, during the interface flap at the peer device, the vMX occasionally stops receiving traffic. FPC needs to be restarted to recover functionality on the affected interface. [PR1401672](#)
- The **/etc/passwd** file is created during the first commit when a new jinstall image is used to boot for the first time. If you configure **event-options**, the system tries to read the configuration from the available event scripts, which requires privileges obtained from the **/etc/passwd** file. This creates a dependency as the commit will not be successful if the configuration includes **event-options** the first time a new image boots up, which is the case when an upgrade is performed with the **virsh create** command. [PR1220671](#)
- On vMX instances, the vFPC might stop working if the **family iso** is enabled through GRE over IPsec tunnel. [PR1364624](#)
- The default NIC adapter type has changed from E1000 to VMXNET3. As a result, the default setting on the OVA file for vFPC is not set to the correct NIC driver (VMXNET3), and the vFPC is loaded with the E1000 driver. [PR1365337](#)
- On vMX instances, encryption and decryption do not work because the Packet Forwarding Engine discards the group VPN member established using the authentication method preshared key **ascii-text**. [PR1381316](#)
- Jlock assert causes the kernel to stop because of the ifstate code path. [PR1398320](#)

Upgrade Instructions

You cannot upgrade Junos OS for the vMX router from earlier releases using the **request system software add** command.

You must deploy a new vMX instance using the downloaded software package.

System Requirements

IN THIS SECTION

- [Minimum Hardware and Software Requirements for KVM | 5](#)
- [Minimum Hardware and Software Requirements for VMware | 8](#)

These topics provide system requirements for each supported environment.

Minimum Hardware and Software Requirements for KVM

[Table 1 on page 5](#) lists the hardware requirements for KVM.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Required CPUs and NICs	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type Intel X520 NICs using ixgbe driver or Intel X710 using i40e driver are required. Only 10G ports are supported. OEM versions of those NICs are also supported when they use the same drivers.

Table 1: Minimum Hardware Requirements for KVM (*continued*)

Description	Value
Number of cores NOTE: Performance mode is the default mode and the minimum value is based on one port.	For lite mode with lab simulation use case applications: Minimum of 3 • 1 for VCP • 2 for VFP NOTE: If you want to use lite mode when you are running more than 3 vCPUs for the VFP, you must explicitly configure lite mode. For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. NOTE: To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—$(4 * \text{number-of-ports}) + 4$ • With CoS—$(5 * \text{number-of-ports}) + 4$ NOTE: All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.
Memory NOTE: Performance mode is the default mode.	For lite mode: Minimum of 3 GB • 1 GB for VCP • 2 GB for VFP For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended for 16 GB 4 GB for VCP 12 GB for VFP Additional 2 GB recommended for host OS

Table 1: Minimum Hardware Requirements for KVM (continued)

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage.
Other requirements	Intel VT-d capability Hyperthreading (recommended) AES-NI

[Table 2 on page 7](#) lists the software requirements for Ubuntu.

Table 2: Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 14.04 Ubuntu 16.04.5 LTS (recommended host OS) Linux 4.4.0-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11
Required packages NOTE: Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libgl2.0-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev NOTE: libvirt 1.3.1

[Table 3 on page 7](#) lists the software requirements for Red Hat Enterprise Linux.

Table 3: Software Requirements for Red Hat Enterprise Linux

Description	Value
Operating system	- Red Hat Enterprise Linux 7.2 Linux 3.10.0-327.4.5 - Starting with Junos OS Release 17.3R1 Red Hat Enterprise Linux 7.3 Linux 3.10.0-514.6.2
Virtualization	QEMU-KVM 1.5.3

Table 3: Software Requirements for Red Hat Enterprise Linux (*continued*)

Description	Value
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools
NOTE: SR-IOV requires these packages: kernel-devel gcc	NOTE: libvirt 1.2.17 or later

Table 4 on page 8 lists the software requirements for CentOS.

Table 4: Software Requirements for CentOS

Description	Value
Operating system	CentOS 7.2 Linux 3.10.0-327.22.2
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools NOTE: libvirt 1.2.19 To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17.

Minimum Hardware and Software Requirements for VMware

Table 5 on page 9 lists the hardware requirements for VMware.

Table 5: Minimum Hardware Requirements for VMware

Description	Value
Number of cores NOTE: Performance mode is the default mode and the minimum value is based on one port.	For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. NOTE: To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—$(4 * \text{number-of-ports}) + 4$ • With CoS—$(5 * \text{number-of-ports}) + 4$ NOTE: All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX. For lite mode: Minimum of 3 • 1 for VCP • 2 for VFP NOTE: If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.
Memory NOTE: Performance mode is the default mode.	For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended for 16 GB 4 GB for VCP 12 GB for VFP For lite mode: Minimum of 3 GB • 1 GB for VCP • 2 GB for VFP

Table 5: Minimum Hardware Requirements for VMware (*continued*)

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage.
vNICs	• SR-IOV NOTE: SR-IOV is only supported with Intel Ivy Bridge CPU (or higher) and Intel 82599 or X710 controller based NICs. Any other NIC models are not supported (for example,; Broadcom QLogic 57810). • VMXNET3 NOTE: See Virtual Network Interfaces for vMX for an overview on the types of virtual network interfaces supported by vMX.

[Table 6 on page 10](#) lists the software requirements.

Table 6: Software Requirements for VMware

Description	Value
Hypervisor	VMware ESXi 5.5 (Update 2), 6.0, or 6.5 NOTE: Due to a DPDK version change in Junos OS Release 18.1R1 and later, ESXi 6.5 is the minimum version required to run the vMX router if you are operating in high-bandwidth mode (performance mode). If you are operating the vMX router in low-bandwidth(lite) mode, you can use ESXi 6.0 or ESXi 5.5.
Management Client	vSphere 5.5 or vCenter Server

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or Partner Support Service support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at <https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf>.
- Product warranties—For product warranty information, visit <http://www.juniper.net/support/warranty/>.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: <https://www.juniper.net/customers/support/>
- Search for known bugs: <https://prsearch.juniper.net/>
- Find product documentation: <https://www.juniper.net/documentation/>
- Find solutions and answer questions using our Knowledge Base: <https://kb.juniper.net/>
- Download the latest versions of software and review release notes: <https://www.juniper.net/customers/csc/software/>
- Search technical bulletins for relevant hardware and software notifications: <https://kb.juniper.net/InfoCenter/>
- Join and participate in the Juniper Networks Community Forum: <https://www.juniper.net/company/communities/>
- Create a service request online: <https://myjuniper.juniper.net>

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: <https://entitlementsearch.juniper.net/entitlementsearch/>

Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit <https://myjuniper.juniper.net>.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see <https://support.juniper.net/support/requesting-support/>.

Revision History

21 July 2020—Revision 6, Junos OS Release 18.4R1—vMX

14 March 2019—Revision 5, Junos OS Release 18.4R1—vMX

5 March 2019—Revision 4, Junos OS Release 18.4R1—vMX

27 February 2019—Revision 3, Junos OS Release 18.4R1—vMX

21 February 2019—Revision 2, Junos OS Release 18.4R1—vMX

21 December 2018—Revision 1, Junos OS Release 18.4R1—vMX

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